1. Y d-dim irreducible projective variety over an algebraically closed field $\mathbb{F}$. (projective variety: integral projective scheme over a field $\mathbb{F}$.) $\operatorname{Proj} A[x_0, \dots, x_n] = \mathbb{P}^n_A$.

2. intersection product $\operatorname{Div}(Y)^d \longrightarrow \mathbb{Z}$ ✓

4.2. **Projective varieties and Lorentzian polynomials.** Let Y be a d-dimensional irreducible projective variety over an algebraically closed field $\mathbb{F}$. If $\mathrm{D}_1, \dots, \mathrm{D}_d$ are Cartier divisors on Y, the *intersection product* $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ is an integer defined by the following properties:

– the product $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ is symmetric and multilinear as a function of its arguments,

– the product $(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d)$ depends only on the linear equivalence classes of the D_i, and

– if $\mathrm{D}_1, \dots, \mathrm{D}_n$ are effective divisors meeting transversely at smooth points of Y, then

$$(\mathrm{D}_1 \cdot \ldots \cdot \mathrm{D}_d) = \#\mathrm{D}_1 \cap \ldots \cap \mathrm{D}_d.$$

$\operatorname{Pic}(Y) \cong \operatorname{Div}(Y) /$ linearly equivalence.

$\rightsquigarrow$ $\mathbb{Q}$-divisors intersection product $\mathrm{Div}_{\mathbb{Q}}(Y)^d \longrightarrow \mathbb{Q}$.

3. volume polynomial $\mathrm{vol}_H(\omega) = (\omega_1 H_1 + \cdots + \omega_n H_n)^d$, H_i $\mathbb{Q}$-divisors.

main theorem: If $H_1 \cdots H_n$ are nef divisors, $\mathrm{vol}_H(\omega)$ is a Lorentzian polynomial.

all the Cartier divisors.

4. $\mathrm{Div}(Y)$ is a group.

$D_1 \sim D_2$ if $(D_1 \cdot C) = (D_2 \cdot C)$ $\forall$ irreducible curve C.

$\mathrm{Div}(Y) / \text{numerical equivalence} \cong \mathbb{Z}^{\rho}$ a f.g. abelian group.

discrete topology.

(ρ is called Picard number of Y)

all the $\mathbb{Q}$-divisors

$$\mathrm{Div}_{\mathbb{Q}}(Y)/\text{numerical equivalence} \cong \left(\mathrm{Div}(Y)/\text{numerical equivalence}\right) \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Z}^{\rho} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}^{\rho}$$

finite-dim $\mathbb{Q}$-vector space

$N^1(Y)_{\mathbb{Q}}$

$\rightsquigarrow$ Euclidean topology.

In this top, every nef divisors is a limit of ample divisors.

invertible sheaf $\rightsquigarrow$ ample divisor. ✓ ← 上次

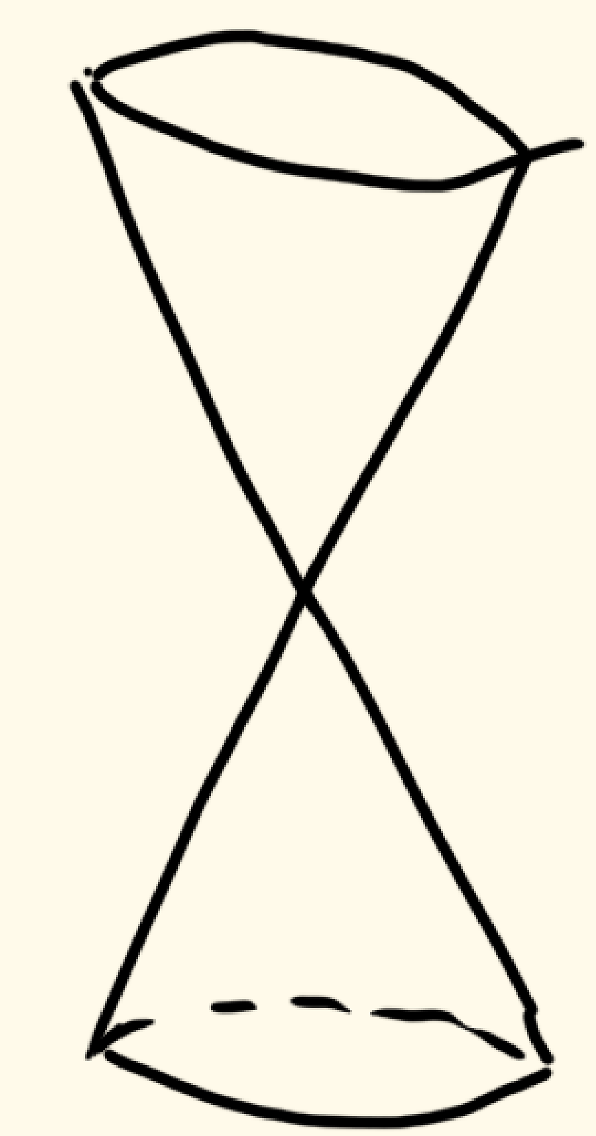
Cartier divisor D is very ample if there is a closed immersion

$i : Y \hookrightarrow \mathbb{P}^n_k$ s.t. $i^*(\mathcal{O}_{\mathbb{P}^n_k}(1)) \cong \mathcal{O}_Y(D)$.

$\mathbb{Q}$-divisor D is ample if mD is very ample for some $m \in \mathbb{Z}^+$.

nef divisor : $\mathbb{Q}$-divisor D is nef if $(D \cdot C) \geq 0$ for any irr curves $C \subseteq Y$.

$\mathrm{Amp}(Y)$: ample divisors / numerical equivalence is an open cone in $N^1(Y)_{\mathbb{Q}}$.



$D \in \mathrm{Amp}(Y)$, then $\lambda D \in \mathrm{Amp}(Y)$, $\forall \lambda \in \mathbb{Q}$.

$\mathrm{Nef}(Y)$: nef divisors $/ \sim$ is closed cone.

Moreover, $\mathrm{Nef}(Y) = \overline{\mathrm{Amp}(Y)}$.

Proof. Step 1. It suffices to prove this for all very ample divisors. ($H_1, \cdots, H_n$ very ample).

Now assume H_i are very ample.

Step 2. By Nakai-Moishezon criterion,

$V_\alpha(H) = (H_1^{\alpha_1} \cdots H_n^{\alpha_n})$, $\alpha \in \Delta_n^d$

If H_i are ample, then $V_\alpha(H) > 0$.

$$\mathrm{vol}_H(\omega) = (\omega_1 H_1 + \cdots + \omega_n H_n)^d = \sum_{\alpha \in \Delta_n^d} \frac{d!}{\alpha!} V_\alpha(H)\, \omega^\alpha$$

Step 3. L_n^d is the closure of $\mathring{L}_n^d$ in H_n^d.

$\rightsquigarrow$ It is enough to show $\partial^\alpha \mathrm{vol}_H \in \mathring{L}_n^2$, $\forall \alpha \in \Delta_n^{d-2}$.

多项式只有正特征值.

$\forall \alpha \in \Delta_n^{d-2}$, $\frac{2!}{d!} \partial^\alpha \mathrm{vol}_H(w) = \left(\sum_{i=1}^n w_i H_i \cdot \sum_{i=1}^n w_i H_i \cdot H_1^{\alpha_1} \cdots H_n^{\alpha_n} \right)$.

$(w_1 H_1 + \cdots + w_n H_n)^d$ 求$(d-2)$次导.

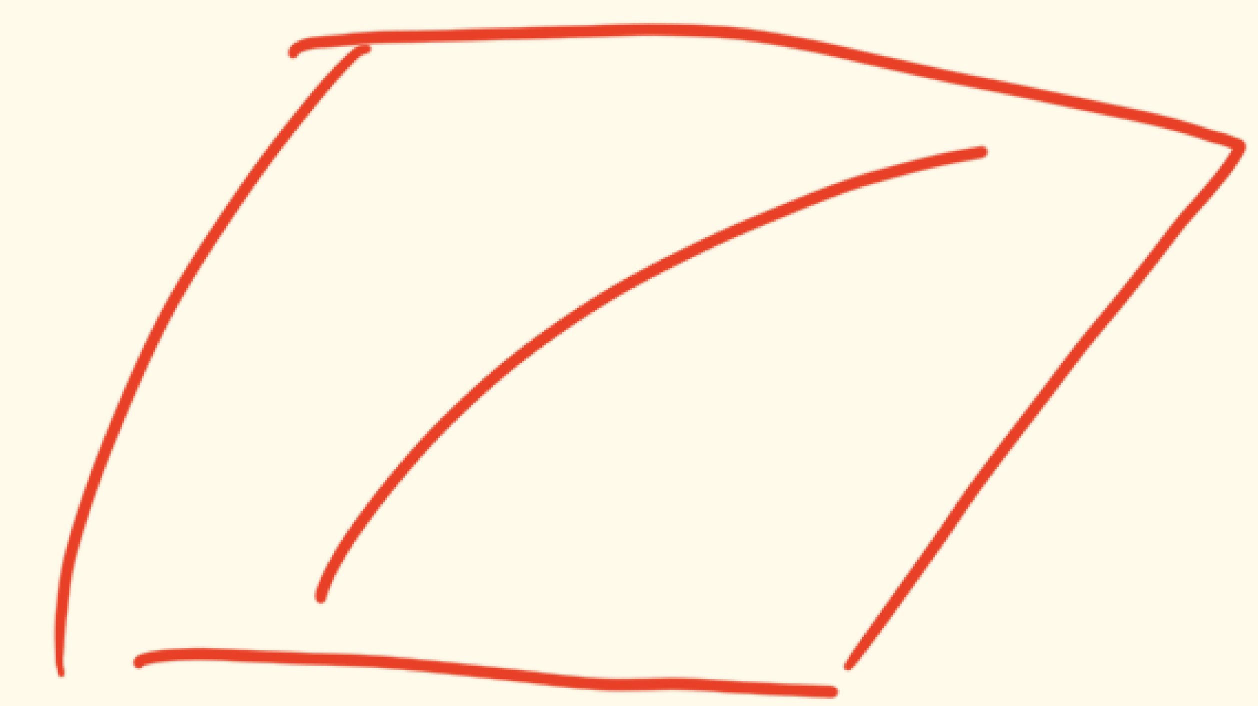
Step 4. $|H| := \{ D \in \mathrm{Div}(Y) : D$ is effective $(D \geq 0)$, D is linearly equivalent to $H \}$, $H \in \mathrm{Div}(Y)$

linear system.

H is very ample, $|H|$ hyperplane sections of Y.

Geometric meaning: H is very ample, $L \subseteq \mathbb{P}^N_F$ is a hyperplane, then $Y \cap L$ is subvariety.

Bertini's theorem, $L \in |H|$ is an irreducible subvariety of dim $d-1$, H is very ample.



$$\left(\sum_{i=1}^{n} w_i H_i \cdot \sum_{i=1}^{n} w_i H_i \cdot H_1^{\alpha_1} \cdots H_n^{\alpha_n} \right). \;\star$$

Pick a hyperplane $H_1^{(gen)} \in |H_1|$, an irr $d-1$ subvariety.

effective divisor $\xleftrightarrow{1:1}$ subscheme

$H_2^{(gen)} \in |H_2|$ over the new variety, $\rightsquigarrow$ irr $d-2$ subvariety.

$d-2$ steps

$\Longrightarrow$ an irreducible surface S.

Intersection theory

$$\left(\sum_{i=1}^{n} w_i H_i \cdot \sum_{i=1}^{n} w_i H_i \cdot H_1^{\alpha_1} \cdots H_n^{\alpha_n}\right) \text{ on } Y$$

$$= \left(\sum_{i=1}^{n} w_i H_i \cdot \sum_{i=1}^{n} w_i H_i\right) \text{ on } S.$$

irreducible surface.

Step 5. Hodge's index theorem (on a surface).

Hodge Index Theorem:
Let S be a smooth, irreducible projective surface, and let H be an **ample** divisor on S.

- If D is any divisor class on S such that $D \cdot H = 0$ (meaning D is orthogonal to the ample class),
- Then $D^2 \leq 0$ (the self-intersection of D must be negative or zero).
- Furthermore, $D^2 = 0$ if and only if D is numerically equivalent to zero.

$\left(\sum_{i=1}^{n} w_i H_i \cdot \sum_{i=1}^{n} w_i H_i \right)$ on S. $\rightsquigarrow$ strictly Lorentzian.

has exactly one positive eigenvalue.

I. $\boxed{S \text{ is smooth,}}$

Fix $H_i|_S$.

$(H_i \cdot H_i)_S > 0$ take $D \in \mathrm{Div}(S)$. s.t. $D \cdot H = 0$.

$$N^1(S)_{\mathbb{Q}} = (\mathbb{Q} \cdot H_i) \oplus H^{\perp}$$

$$\uparrow$$

$$(D \cdot H_i) = 0$$

$\rightsquigarrow (D \cdot D)_S \leq 0.$ exactly one positive eigenvalue.

2. S is not smooth. Projection Formula.

S is singular. $\exists !\ \tilde{S}$ s.t. $\tilde{S}$ is smooth and projective and $\exists$ a proper birational morphism $\pi : \tilde{S} \to S$.

D_1, D_2 is Cartier divisor, $(\pi^* D_1 \cdot \pi^* D_2)_{\tilde{S}} = (D_1 \cdot D_2)_S$.

$$
\begin{pmatrix}
(H_1 \cdot H_1)_S & \cdots & (H_1 \cdot H_n)_S \\
\vdots & & \vdots \\
(H_n \cdot H_1)_S & \cdots & (H_n \cdot H_n)_S
\end{pmatrix}
$$

$$\parallel$$

$$
\begin{pmatrix}
(\pi^* H_1 \cdot \pi^* H_n)_{\tilde{S}} & \cdots \\
\vdots & \vdots \\
\cdots &
\end{pmatrix}
\longleftarrow \text{exactly one positive eigenvalue.}
$$

$\blacksquare$.